

Clinical Study Summary Report (CSSR)

Document Status: [Draft / Final] | **Version:** [1.0] | **Date:** [DD-MMM-YYYY]

1. Administrative & Study Identification

Full Official Study Title: A Three-part, Phase I/II Study to Investigate the Safety, Tolerability, Pharmacokinetics, and Efficacy of Zifibancimig Following Intravitreal Administration of Multiple Ascending Doses and Continuous Delivery From the Port Delivery in Patients With Neovascular Age-related Macular Degeneration **Brief Title:** Study of Zifibancimig in Participants With Neovascular Age-related Macular Degeneration **Short Title/ Acronym:** BURGUNDY **Protocol Number:** BP41670 **NCT Number:** NCT04567303 **Posting ID:** 947170500794734699 **Sponsor Name:** Roche (Hoffmann-La Roche) **Investigational Product:** Zifibancimig (R07250284) and Ranibizumab (Trade names: Lucentis, Susvimo, Cimerli, Byooviz) via Port Delivery Platform (PDP) **Phase of Development:** Phase I/II (PHASE1, PHASE2) **Trial Status:** Recruiting **Medical Monitor:** Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective

*Primary Objective: This is a first-in-human study to investigate the safety, tolerability, and efficacy of zifibancimig administered through intravitreal (IVT) injections and via the port delivery (PD) implant in participants with neovascular age-related macular degeneration (nAMD). * Secondary Objectives:** To evaluate the pharmacokinetic (PK) profile of zifibancimig in blood and aqueous humor, measure anatomic and visual acuity changes, and assess the functionality and refill intervals of the port delivery platform.

2.2 Background & Rationale

To address the high treatment burden of repeated intravitreal injections in patients with nAMD (Macular Degeneration). The port delivery system (PDS) allows for continuous, sustained delivery of anti-VEGF medications into the vitreous. This study explores the novel bispecific antibody zifibancimig (targeting both angiopoietin-2 and VEGF) in continuous delivery format, aiming to maintain visual gains and anatomical improvements while significantly reducing the frequency of clinic visits and injections required for standard nAMD management.

3. Study Design & Methodology

3.1 Design Framework

*Study Type: Interventional * Control Type: Active Comparator (Ranibizumab) / Dose Comparison (High vs. Low dose) * Blinding: Partially Masked (Unmasked for multiple ascending doses; Masked*

sustained delivery) * *Randomization:** Randomized (1:1 for the dose levels in the PD implant)

3.2 Planned Enrollment & Duration

Targeted Enrollment (\$N\$): 251 participants

Number of Sites: Multicenter global study

Estimated Study Start: 28-Oct-2020

Estimated Primary Completion: 01-Aug-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria

1. Willing to allow Aqueous Humor (AH) collection. 2. Choroidal neovascularization (CNV) exclusively due to age-related macular degeneration (AMD). 3. Treatment-naïve or pre-treated with anti-VEGF/Ang-2 (last IVT administration must have occurred at least 21 days prior to the screening visit). 4. Diagnosis of nAMD within 36-months prior to the screening visit. 5. Decreased Best-Corrected Visual Acuity (BCVA) attributable primarily to nAMD, with a BCVA letter score of 78 to 34 letters (inclusive) on ETDRS-like charts at screening.

4.2 Exclusion Criteria

1. CNV due to causes other than nAMD, such as ocular histoplasmosis, pathological myopia, trauma, angioid streaks, choroidal rupture, uveitis, or central serous chorioretinopathy. 2. Actual or history of myopia >-8 diopters. 3. History of vitrectomy surgery, submacular surgery, other intraocular surgery, or any planned surgical intervention during the study period. 4. Prior participation in a clinical trial involving anti-VEGF drugs within six months (other than ranibizumab, aflibercept, or faricimab). Prior treatment with brodalumab or prior gene therapy for nAMD. 5. Active intraocular inflammation (grade trace or above), infectious conjunctivitis, keratitis, scleritis, endophthalmitis, or history of uveitis. 6. Treatment with IVT anti-VEGF or anti-VEGF/Ang-2 agents within one week prior to Day 1 in the fellow eye (concurrent treatment with SUSVIMO™ in the fellow eye is not exclusionary).

5. Intervention & Treatment Plan

5.1 Investigational Regimen

Dosage/Frequency:

* *Part 1: Multiple ascending doses of zifibancimig (R07250284) by intravitreal (IVT) injection at a constant volume of 50 μ L.* * *Part 2 & 3: Randomized to either a low or high dose of zifibancimig administered through the surgically implanted port delivery (PD) system, followed by continuous delivery and subsequent refill-exchanges with ranibizumab via the PD implant.*

* *Route of Administration: Intravitreal (IVT) injection and surgical Port Delivery (PD) Implant.* * *Duration of Treatment:** Long-term continuous extended release, up to study completion in 2026.

5.2 Concomitant & Prohibited Therapy

*Permitted: Standard concomitant therapies for systemic comorbidities; transition to ranibizumab PD implant refills as indicated in the protocol. Concurrent SUSVIMO™ treatment in the fellow eye is permitted. * Prohibited:* Brolucizumab, any anti-VEGF-C or anti-VEGF-D inhibitors, or treatment with any medication for geographic atrophy (GA) during the prior 3 months.

6. Study Timeline & Visit Schedule

- **Screening:** Day -28 to 0 (Informed Consent, Historical VA data assessment, SD-OCT/FA imaging, Med History).
- **Baseline:** Day 1 (Randomization, Part 1 first IVT dose, or surgical implantation of the PD system).
- **Interim:** Every 4 to 12 weeks (Safety exams, BCVA measurements, continuous delivery monitoring, refill-exchange procedures if indicated).
- **End of Study:** Final assessments, explantation (if applicable) or transition to standard of care, exit interview.

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes

1. Percentage of Participants With Ocular and Systemic (Non-ocular) Adverse Events (AEs). 2. Percentage of Participants With Ocular AEs During the Post-operative and Follow-up Periods. 3. Percentage of Participants With Adverse Events of Special Interest (AESIs) Including Ocular AESIs.

7.2 Secondary Outcomes

1. Maximum Observed Concentration (C_{max}) of Zifibancimig in Blood and Aqueous Humor (AH). 2. Time of Maximum Concentration Observed (T_{max}) of Zifibancimig in Blood and AH. 3. Concentration at the End of a Dosing Interval Before the Next Dose Administration (C_{trough}) of Zifibancimig in Blood and AH. (*Exploratory metrics also include change from baseline in BCVA using the ETDRS scale and central macular thickness via SD-OCT*).

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Continuous monitoring of both ocular AEs (e.g., conjunctival hyperemia, endophthalmitis, cataract, intraocular inflammation) and systemic AEs. Tracking of post-operative period specific occurrences.
- **Serious Adverse Events (SAEs) & AESIs:** Protocol-mandated rapid reporting specifically tracking adverse events of special interest related to the surgical implantation of the PDS.
- **Data Monitoring Committee (DMC):** Independent safety monitoring committee overseeing the dose-escalation (Part 1) and PD implant phases (Parts 2 & 3).

9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Intent-to-Treat (ITT) for primary efficacy; Safety Set for all AE/SAE evaluations.
- **Sample Size Justification:** Targeted enrollment of N=251 provides adequate power to characterize the PK profile and capture safety signals of multiple ascending doses and PD delivery.
- **Handling of Missing Data:** Mixed-Model Repeated Measures (MMRM) and Last Observation Carried Forward (LOCF) for BCVA and central thickness endpoints.

10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
- **Monitoring Plan:** Frequent onsite monitoring to oversee surgical implantation safety and correct handling of the Port Delivery Platform.
- **Audit Trail:** Standard tracking of eCRF, centralized reading center documentation for SD-OCT and FA imaging, and PK sampling chain-of-custody.

11. Study Identifiers & Governance

- **Primary ID:** BP41670
- **Registry Number:** NCT04567303
- **Posting ID:** 947170500794734699
- **Sponsor/Lead Organization:** Roche
- **Brief Title:** Study of Zifibancimig in Participants With Neovascular Age-related Macular Degeneration
- **Official Regulatory Title:** A Three-part, Phase I/II Study to Investigate the Safety, Tolerability, Pharmacokinetics, and Efficacy of Zifibancimig Following Intravitreal Administration of Multiple Ascending Doses and Continuous Delivery From the Port Delivery in Patients With Neovascular Age-related Macular Degeneration
- **Status:** Recruiting

12. Clinical Framework (nAMD Specific)

- **Primary Condition:** Neovascular Age-related Macular Degeneration (nAMD) / Macular Degeneration
- **Diagnosis Threshold:** Active CNV exclusively due to AMD confirmed by SD-OCT and FA, with vision loss primarily attributable to nAMD.
- **Medical Methodology:** Anti-VEGF / Anti-Ang-2 therapies and continuous surgical Port Delivery (PD) Platform.
- **Optimization Target:** Reduction in treatment/injection burden while maintaining/improving visual acuity and retinal thickness.

13. Study Procedures & Methodology

- **Management Pathway:** Continuous release of drug via the Port Delivery Platform with scheduled refill-exchanges.
- **Education Protocol (HCP):** Specialized surgical training for the pars plana incision and proper PDS implantation/explantation/refill techniques.
- **Education Protocol (Patient):** Post-operative care guidelines, recognition of endophthalmitis/inflammation signs, and maintaining follow-up adherence.
- **Quality Audit Mechanism:** Central reading center for masked image grading (SD-OCT, FA, FP, FAF).

14. Observation & Follow-Up Schedule

- **Total Duration:** Up to approximately 6 years (Estimated completion 2026).
- **Onsite Visit Cadence:** Baseline, frequent immediate post-surgical follow-ups, and interval visits mapped to continuous delivery refill-exchanges.
- **Remote Monitoring Frequency:** Standard check-ins per site standard of care for AEs.
- **Checklist Review Interval:** Regular tracking of visual acuity (ETDRS), anatomic measurements, and PK sample collections from Blood/AH.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				